

Ultraviolet phototoxicity to the retina.

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OBJECTIVE: This overview of ultraviolet (UV) phototoxicity considers the interaction of UVA and short-wavelength VIS light with the retina and retinal pigment epithelium. METHODS: The damage mechanisms underlying UV retinal phototoxicity are illustrated with a literature survey and presentation of experimental results. RESULTS: Depending on the wavelength and exposure duration, light interacts with tissue by three general mechanisms: thermal, mechanical, or photochemical. Although the anterior structures of the eye absorb much of the UV component of the optical radiation spectrum, a portion of the UVA band (315-400 nm) penetrates into the retina. Natural sources, such as the sun, emit energetic UV photons in relatively long durations, which typically do not result in energy confinement in the retina, and thus do not produce thermal or mechanical damage but are capable of inducing photochemical damage. Photochemical damage in the retina proceeds through Type 1 (direct reactions involving proton or electron transfers) and Type 2 (reactions involving reactive oxygen species) mechanisms. Commonly used drugs, such as certain antibiotics, nonsteroidal anti-inflammatory drugs, psychotherapeutic agents, and even herbal medicines, may act as photosensitizers that promote retinal UV damage, if they are excited by UVA or visible light and have sufficient retinal penetration. CONCLUSIONS: Although the anterior portion of the eye is the most susceptible to UV damage, the retina is at risk to the longer UV wavelengths that propagate through the ocular media. Some phototoxicity may be counteracted or reduced by dietary intake of antioxidants and protective phytonutrients.